

Unstructured Mesh Generation

Technical Progress Report (Week 11)

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Week 11 Focus: Stability, UX & Correctness

Objective

No new features added this week. Full focus on **improving robustness, fixing bugs, and enhancing the user experience** of existing tools.

Key improvement areas:

- ◀ Geometric validation and smart error messages
- ◀ Performance optimization (no UI freezes)
- ◀ Metric unit display on all length inputs
- ◀ Stable multi-frame CFD animation pipeline

1. Geometric Validation for Trailing Edge

Problem: Trailing Edge Refinement was silently applied to all objects, including spheres – where it is physically meaningless.

Solution: bmesh Edge Scan

- ◀ Scans all edges using the bmesh API before meshing
- ◀ Computes face-face angle for each edge
- ◀ If **no edge exceeds 30°**: shows a clear error message
- ◀ Allows objects with **both curved and sharp edges** (e.g. cylinders with rims)

“Trailing edge refinement requires sharp edges. Your geometry appears completely smooth (e.g. a sphere). Please disable this feature.”

2. Non-Manifold Geometry Pre-Check

Problem: cfMesh silently fails or produces corrupt meshes when the input STL has holes or self-intersections.

Solution: Pre-Meshing Validation

- ◀ bmesh scan detects non-manifold edges before export
- ◀ Reports exact edge count: *"X non-manifold edges detected – please fix in Edit Mode"*
- ◀ Non-blocking warning (cfMesh may still succeed on simple cases)

3. Performance: Triangulation Optimization

Problem: Previous implementation used `bpy.ops.object.modifier_add()` in a loop, triggering a full Blender dependency graph rebuild per object. This caused **UI freezes** on complex meshes.

Solution: Low-Level bmesh Operations

- ◀ Replaced with `bmesh.ops.triangulate(bm, faces=bm.faces[:])`
- ◀ Operates directly on mesh data – no graph rebuilds
- ◀ Significantly faster on dense meshes

Method	Graph Rebuild	Speed
<code>bpy.ops</code> loop	Per object	Slow
<code>bmesh.ops</code>	None	Fast

4. Beginner-Friendly Error Messages

Problem: Errors surfaced raw Python exceptions, which are meaningless to non-developers.

Improved Exception Handling (3 files)

- ▶ **ops_meshing.py:** `PermissionError` → “Check write permissions”; `MemoryError` → “Increase cell size”
- ▶ **ops_geometry.py:** STL import errors now guide users to valid ASCII/Binary STL files
- ▶ **ops_inspect.py:** VTU parse error now says “Did you run the mesher first?”
- ▶ **OpenFOAM guard:** Checks if `cartesianMesh` is accessible before starting any operation

5. Metric Units & Live Estimators in UI

Metric Units

- ◀ Added subtype='DISTANCE' to all length-based properties
- ◀ Blender now auto-displays "m", "cm", "mm" next to inputs
- ◀ Applies to: cell size, layer thickness, characteristic length, trailing edge size

Live Cell Count Estimator

- ◀ Before meshing: displays "Est. Cells: $\sim 5,000$ "
- ◀ Formula: $n \approx V_{\text{bbox}}/dx^3$
- ◀ Warns in red if $> 2 \times 10^6$ cells
- ◀ Helps users avoid out-of-memory crashes

6. Courant Number Warning System

Problem: icoFoam diverges when the Courant number $Co = U \cdot \Delta t / \Delta x > 1$. Users had no feedback before running.

Live Courant Checker (Physics & Solver Panel)

$$Co = \frac{U_{\text{mag}} \cdot \Delta t}{\Delta x}, \quad \Delta t_{\text{safe}} = \frac{0.5 \cdot \Delta x}{U_{\text{mag}}}$$

- ◀ Updates live as you change velocity, Δt , or cell size
- ◀ **Green:** $Co < 0.5$ (stable)
- ◀ **Orange:** $0.5 < Co < 1.0$ (borderline)
- ◀ **Red alert:** $Co \geq 1$ (solver WILL diverge) + shows safe Δt

7. Estimated Animation Frames in Time Controls

Problem: Users did not know how many output files would be written before running a simulation.

Live Frame Count Preview

icoFoam (adjustableRunTime):

$$n_{\text{frames}} = \text{write_interval} + 1$$

simpleFoam (step-based):

$$n_{\text{frames}} = \lfloor \text{max_iterations} / \text{write_interval} \rfloor$$

- ◀ Shown live inside the Time & Interval box
- ◀ Warns if 0 or 1 frame (too coarse interval)
- ◀ Warns if > 500 frames (disk space risk)

8. Animation Pipeline: Multi-Frame Fix

Problem: Animation showed only 1 visible frame regardless of time steps.

Root Cause

Old keyframe logic set each mesh visible for exactly 1 Blender frame, then immediately hidden again: *frame N-1: hidden, frame N: visible, frame N+1: hidden*

Fix

- ◀ Each mesh visible from its frame until the **next one appears**
- ◀ CONSTANT keyframe interpolation (no blending between steps)
- ◀ Compatible with Blender 3.x / 4.x / 5.x action API
- ◀ Last frame stays visible permanently

9. Parallel Run Reconstruction (reconstructPar)

Problem: When using > 1 CPU cores, OpenFOAM writes results to `processor0/`, `processor1/...`, **not** the main case directory. `foamToVTK` saw only the initial condition, producing just 1–2 frames.

Automatic Fix in “Color Boundary Mesh”

- 1 Detects `processor*` directories automatically
- 2 Runs `reconstructPar -newTimes` before `foamToVTK`
- 3 Only merges new time steps (fast on re-run)
- 4 Falls back gracefully with a warning if reconstruction fails

Result: 21 frames correctly generated from 20 write intervals.

10. Other Post-Processing Improvements

- ◀ **Auto Shade Smooth:** Imported result STL is automatically smooth-shaded, eliminating the “faceted sphere” appearance without manual steps
- ◀ **Mesh Quality Icons:** Non-ortho, skewness, and aspect ratio fields each display ✓ / ○ / ✗ icons based on OpenFOAM thresholds
- ◀ **Octree Exact-Multiple Bug:** maxCellSize is nudged by 10^{-5} relative epsilon before writing meshDict, preventing the “0 cells in mesh” crash
- ◀ **Live Log Tail:** Status box updates in real-time with the last cfMesh/OpenFOAM output line during execution

Updated Timeline & Milestones

Week	Task	Status
4	Turbulence models (RANS), Exception handling	Done
5 – 6	STL Importer, Jet Colormaps, Residuals & Y+	Done
7	System file opener, Custom liquid presets & Smart UI	Done
8	OpenFOAM 2412 migration, Modular Architecture	Done
9 – 10	Real-time Vis, VTK Slices, icoFoam & simpleFoam	Done
11	Stability, UX Polish, Animation, Bug Fixes	Done
12	Final Documentation & Project Handover	Upcoming

Week 11 Summary

Improvement	Outcome
Geometric Validation	Trailing edge only on sharp-edged geometry
Non-Manifold Check	Warns before meshing broken STL
Triangulation Speed	No UI freeze on large meshes
Beginner Error Messages	Plain-English guidance instead of Python tracebacks
Metric Units	"m" / "mm" shown on all distance inputs
Courant Warning	Live Co number; prevents icoFoam divergence
Frame Count Estimator	Accurate preview in Time & Interval panel
Animation Fix	All time steps visible; CONSTANT keyframes
reconstructPar	Parallel runs now produce full animation

Thank You!

End of Week 11 Report